# REVIEW: EXAM 3

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Name: Solutions

**1.** Let

$$f(x) = \int_0^x \cos(t)\,\mathrm{d}t.$$

Use Part I of the Fundamental Theorem of Calculus to evaluate

$$\frac{\mathrm{d}}{\mathrm{d}x} f(x^2) = \frac{\mathrm{d}}{\mathrm{d}x} \int_0^{x^2} \cos(t)\,\mathrm{d}t.$$

$f'(x) = \cos(x)$

$\frac{d}{dx} f(x^2) = f'(x^2)\frac{d}{dx}x^2 = 2x f'(x^2) = \boxed{2x\cos(x^2).}$

**2.** Evaluate the integral $\int_0^2 3x^2\,dx$ using the limit definition of the integral and the identity

$$\sum_{i=1}^{n} i^2 = \frac{n(n+1)(2n+1)}{6} = \frac{2n^3+3n^2+n}{6}.$$

$f(x) = 3x^2, \quad \Delta x = \frac{2-0}{n} = \frac{2}{n}, \quad x_i = 0 + i\Delta x = \frac{2i}{n}$

$f(x_i) = 3\left(\frac{2i}{n}\right)^2 = \frac{3(4)}{n^2} i^2 = \frac{12}{n^2} i^2$

$R_n = \sum_{i=1}^{n} f(x_i)\Delta x = \sum_{i=1}^{n} \frac{12}{n^2} i^2 \left(\frac{2}{n}\right) = \frac{24}{n^3} \sum_{i=1}^{n} i^2 = \frac{24}{n^3}\left(\frac{2n^3+3n^2+n}{6}\right) = 4\left(\frac{2n^3}{n^3} + \frac{3n^2}{n^3} + \frac{n}{n^3}\right) = 4\left(2 + 3/n + 1/n^2\right)$

$\int_0^2 3x^2\,dx = \lim_{n\to\infty} R_n = 4(2+0+0) = \boxed{8.}$

**3.** Use the Fundamental Theorem of Calculus Part II to check your answer to Problem 2.

$\int_0^2 3x^2\,dx = 3\left(\frac{1}{3}\right)x^3\Big|_0^2 = 2^3 - 0^2 = \boxed{8.}$

**4.** Evaluate the indefinite integral

$$\int \frac{\sin(x)}{\cos^2(x)}\,\mathrm{d}x.$$

$$\int \frac{\sin(x)}{\cos^2(x)}dx = \int \frac{\sin(x)}{\cos(x)}\frac{1}{\cos(x)}dx = \int \tan(x)\sec(x)\,dx = \boxed{\sec(x)+C.}$$

**5.** Evaluate the indefinite integral

$$\int \csc(\theta)\left(\sin(\theta) - \csc(\theta)\right)\mathrm{d}\theta$$

$$\int \csc(\theta)\sin(\theta)d\theta + \int -\csc^2(\theta)d\theta = \int \frac{\sin(\theta)}{\sin(\theta)}d\theta + \int -\csc^2(\theta)d\theta$$

$$= \int d\theta + \int -\csc^2(\theta)d\theta$$

$$= \boxed{\theta + \cot(\theta) + C}$$

**6.** Evaluate the indefinite integral

$$\int 2x\sqrt{x-5}\,\mathrm{d}x$$

$u = x-5 \iff u+5 = x$

$du = dx$

$$\begin{aligned}\int 2x\sqrt{x-5}\,dx &= \int 2(u+5)\sqrt{u}\,du \\ &= \int 2u\sqrt{u}\,du + \int 10\sqrt{u}\,du \\ &= 2\int u^{3/2}du + 10\int u^{1/2}du \\ &= 2\left(\tfrac{2}{5}u^{5/2}\right) + 10\left(\tfrac{2}{3}\right)u^{3/2} + C \\ &= \boxed{\frac{4}{5}(x-5)^{5/2} + \frac{20}{3}(x-5)^{3/2} + C.}\end{aligned}$$

**7.** Evaluate the definite integral

$$\int_0^{\sqrt{\pi}} x\sin(x^2)\,\mathrm{d}x$$

$u = x^2$

$du = 2x\,dx$

$\Rightarrow \frac{1}{2}du = x\,dx$

$$\begin{aligned}\int_0^{\sqrt{\pi}} x\sin(x^2)\,dx &= \frac{1}{2}\int_0^{\pi}\sin(u)\,du \\ &= -\frac{1}{2}\cos(u)\Big|_0^{\pi} \\ &= -\frac{1}{2}\left(\cos(\pi) - \cos(0)\right) \\ &= -\frac{1}{2}(0-1) = \boxed{\frac{1}{2}}\end{aligned}$$

**8.** Given $f'(x) = 12x^2 + 6x - 4$ and $f(1) = 1$, find $f(x)$.

$f(x) = \int 12x^2+6x-4\,dx = 12\frac{1}{3}x^3+6\frac{1}{2}x^2-4x+C = 4x^3+3x^2-4x+C$

$1 = f(1) = 4+3-4+C = 3+C \Rightarrow C = 1-3 = -2$

$\boxed{f(x) = 4x^3+3x^2-4x-2.}$

**9.** Assume that $f$ is an even function. Given

$$\int_{-1}^{3} f(x)\,\mathrm{d}x = 3 \text{ and } \int_{1}^{3} f(x)\,\mathrm{d}x = 1$$

find

$$\int_{0}^{1} f(x)\,\mathrm{d}x.$$

$\int_{-1}^{3} f(x)dx = \int_{-1}^{1} f(x)dx + \int_{1}^{3} f(x)dx = 2\int_{0}^{1} f(x)dx + \int_{1}^{3} f(x)dx$

$\Rightarrow 2\int_{0}^{1} f(x)dx = \int_{-1}^{3} f(x)dx - \int_{1}^{3} f(x)dx = 3-1=2$

$\boxed{\Rightarrow \int_{0}^{1} f(x)dx = \frac{1}{2}(2) = 1.}$